

FULL STUDY GUIDE: E-COMMERCE SALES PERFORMANCE

Mastering Pandas for Business Decisions (SwiftCart Retail)

Company: SwiftCart Retail

Role: Sales Operations Analyst

Goal

Identify the company's **Cash Cows**, **Top Products**, and **Underperformers** to optimize inventory, marketing, and business decisions for the next quarter.

PHASE 1: LOADING THE DATA & EXPLORATORY DATA ANALYSIS (EDA)

Before making business decisions, we must understand the data structure, quality, and contents.

Step 1: Load the Dataset

1. import pandas as pd
- 2.
3. # Load the dataset
4. df = pd.read_csv("/content/ecommerce_sales_dataset.csv")
- 5.
6. # Display the dataset
7. df

Step 2: Understand the Features (Columns)

8. `df.columns`

Output

```
9. Index(['Order_ID',  
10.      'Product_Category',  
11.      'Price',  
12.      'Quantity',  
13.      'Customer_Rating'],  
14.      dtype='object')
```

Feature Explanation

Column	Description
Order_ID	Unique order identifier
Product_Category	Product category sold
Price	Product price
Quantity	Number of units sold
Customer_Rating	Customer satisfaction score

Decision-Making Insight

Knowing the available columns tells us exactly what business metrics we can calculate:

- Revenue
- Sales Performance
- Customer Satisfaction

- Category Rankings
 - Premium Product Analysis
-

Step 3: View First 5 Rows

15. `df.head()`

Purpose:

Quickly inspect sample records.

Step 4: View Last 5 Rows

16. `df.tail()`

Purpose:

Ensure the dataset loaded correctly from beginning to end.

Step 5: Check Data Types & Missing Values

17. `df.info()`

Example Findings

18. 20 entries

19. No null values

Business Insight

If columns like Quantity or Price contained missing values, revenue calculations would become inaccurate.

Fortunately:

- ✓ No missing values
- ✓ All 20 records available
- ✓ Dataset ready for analysis

Step 6: Generate Statistical Summary

20.df.describe()

Key Insights

Metric	Observation
Average Price	\$2,315.45
Average Rating	3.85
Minimum Rating	2.5
Maximum Price	\$3,890

Business Insight

A rating of 2.5 indicates at least one product is creating customer dissatisfaction and should be investigated.

PHASE 2: BASIC BUSINESS METRICS

These calculations help management understand overall business performance.

Total Orders Placed

```
21. len(df)
```

Output

```
22. 20
```

Interpretation

SwiftCart processed 20 orders.

Number of Product Categories

```
23. df["Product_Category"].nunique()
```

Output

```
24. 4
```

Categories

- Books
 - Clothing
 - Electronics
 - Home
-

Display Category Names

```
25. df["Product_Category"].unique()
```

Output

```
26. ['Books', 'Electronics', 'Clothing', 'Home']
```

Average Product Price

```
27. df["Price"].mean()
```

Output

```
28. 2315.45
```

Interpretation

Average selling price:

\$2,315

This suggests SwiftCart operates primarily in the mid-to-premium market.

Highest Product Price

```
29. df["Price"].max()
```

Output

```
30. 3890
```

Lowest Product Price

```
31. df["Price"].min()
```

Output

PHASE 3: FILTERING DATA (THE "WHERE" CLAUSE)

Filtering allows analysts to investigate specific customer segments and products.

Premium Products (Price > \$3,000)

```
33. df[df["Price"] > 3000]
```

Business Insight

These are high-ticket products.

Recommendation:

Ensure they are always in stock because losing one sale has a significant revenue impact.

Highly Rated Products (Rating > 4)

```
34. df[df["Customer_Rating"] > 4]
```

Business Insight

These products are customer favorites.

Recommendation:

Feature them prominently on:

- Homepage
 - Advertisements
 - Email campaigns
-

Electronics Products Only

```
35. df[df["Product_Category"] == "Electronics"]
```

Purpose:

Analyze the electronics business segment separately.

Count Clothing Orders

```
36. len(df[df["Product_Category"] == "Clothing"])
```

Output

```
37. 4
```

Interpretation

Four orders belong to the Clothing category.

Expensive Electronics

```
38. df[
39.     (df["Product_Category"] == "Electronics")
40.     &
41.     (df["Price"] > 3000)
42.]
```


Business Insight

These are premium technology products.

Recommendation:

Bundle with:

- Extended warranties
 - Insurance plans
 - Accessories
-

Highly Rated Electronics

```
43. df[
44.     (df["Product_Category"] == "Electronics")
45.     &
46.     (df["Customer_Rating"] > 3.5)
47.]
```

Purpose:

Identify electronics customers are satisfied with.

PHASE 4: SORTING DATA

Sorting reveals top and bottom performers immediately.

Sort by Lowest Price

```
48. df.sort_values("Price")
```

Sort by Highest Price

```
49. df.sort_values(  
50.     "Price",  
51.     ascending=False  
52. )
```

Business Insight

Quickly identifies premium products driving revenue.

Sort by Customer Rating

```
53. df.sort_values(  
54.     "Customer_Rating",  
55.     ascending=False  
56. )
```

Business Insight

Allows management to identify:

- Best products (5-star)
- Worst products (2.5-star)

Action:

- Promote high-rated products
 - Investigate low-rated products
-

LEVEL HIGHER! CATEGORY FREQUENCY ANALYSIS

Instead of filtering, count category occurrences.

```
57. df["Product_Category"].value_counts()
```

Output

```
58. Home      7
59. Electronics  6
60. Clothing   4
61. Books      3
```

Business Insight

Home receives the most orders.

Question:

Does Home also generate the most revenue?

Let's find out.

PHASE 5: TASK 1 – REVENUE CALCULATION

Create a revenue column.

Create Total_Sales

```
62. df["Total_Sales"] = (  
63.     df["Price"] * df["Quantity"]  
64. )
```

Formula

65. Revenue = Price × Quantity

Example

Order 1001:

66. Price = \$1,894

67. Quantity = 4

68.

69. Revenue = \$7,576

Average Revenue Per Order

70. `df["Total_Sales"].mean()`

Output

71. $\approx$ \$7,500

Business Insight

The Average Order Value (AOV) is healthy.

PHASE 6: TASK 2 – IDENTIFY TOP PERFORMERS

Use GroupBy to calculate category revenue.

Total Revenue by Category

```
72. category_sales = (  
73.     df.groupby("Product_Category")  
74.     ["Total_Sales"]  
75.     .sum()  
76.     .sort_values(ascending=False)  
77.)  
78.  
79. print(category_sales)
```

Output

```
80. Home      71874  
81. Electronics  38910  
82. Clothing   24963  
83. Books      7576
```

Highest Revenue Category

```
84. top_category = category_sales.idxmax()  
85.  
86. print(  
87.     f"The highest revenue category is: {top_category}"  
88.)
```

Output

```
89. The highest revenue category is: Home
```

Business Insight

Cash Cow

 Home Category

Revenue:

90. \$71,874

Underperformer

 Books Category

Revenue:

91. \$7,576

Recommendation

Inventory Budget Allocation:

- Home → 50%
 - Electronics → 30%
 - Clothing → 15%
 - Books → 5%
-

PHASE 7: TASK 3 – CUSTOMER SATISFACTION ANALYSIS

Measure overall customer happiness.

Average Rating

```
92. avg_rating = (  
93.     df["Customer_Rating"].mean()  
94. )  
95.  
96. print(
```

97. f"Average Customer Rating: {avg_rating:.2f}"
98.)

Output

99. Average Customer Rating: 3.86

Interpretation

Rating Scale:

Rating	Meaning
5	Excellent
4	Good
3	Average
2	Poor

Current Score:

100. 3.86 / 5

Customer satisfaction is moderate.

LEVEL HIGHER! CATEGORY-WISE RATINGS

Find average rating per category.

```
101. df.groupby(  
102.     "Product_Category"  
103. )["Customer_Rating"] \  
104. .mean() \  
105. .sort_values(ascending=False)
```

Output

```
106. Home      4.03  
107. Books     3.80  
108. Electronics 3.77  
109. Clothing  3.65
```

Major Discovery

Home Category:

- ✓ Highest Sales
- ✓ Highest Rating

Business Conclusion

Home products are:

- Most profitable
- Most loved by customers

This is the ideal business segment.

PHASE 8: TASK 4 – PREMIUM PRODUCT FILTERING

Find highly rated products.

Premium Products

Condition:

```
110. Rating > 4

111. premium_products = (
112.     df[df["Customer_Rating"] > 4]
113. )
114.
115. premium_products
```

Count Premium Products

```
116. len(premium_products)
```

Output

```
117. 10
```

Business Insight

These 10 products are SwiftCart's:

★ Golden Products

Recommendation:

- Highlight in ads
 - Include in newsletters
 - Offer loyalty memberships
-

PHASE 9: TASK 5 & 6 – VISUALIZATION

Management prefers visual reports.

TASK 5: BAR CHART

Category vs Revenue

```
118. import matplotlib.pyplot as plt
119.
120. category_sales.plot(
121.     kind="bar",
122.     color="skyblue",
123.     edgecolor="black"
124. )
125.
126. plt.title(
127.     "Total Sales by Product Category",
128.     fontsize=14
129. )
130.
131. plt.xlabel("Product Category")
132. plt.ylabel("Total Sales ($)")
133.
134. plt.xticks(rotation=0)
135.
```

```
136. plt.grid(axis="y", alpha=0.5)
137.
138. plt.show()
```

Chart Analysis

Visual Message:

"HOME IS THE KING"

Home clearly dominates revenue generation.

Business Insight

Management can instantly identify:

- Top category
 - Weak category
 - Resource allocation priorities
-

TASK 6: HISTOGRAM

Product Price Distribution

```
139. df["Price"].hist(
140.     bins=8,
141.     edgecolor="black",
142.     color="orange"
143. )
144.
145. plt.title(
146.     "Distribution of Product Prices",
```

```
147.     fontsize=14
148. )
149.
150. plt.xlabel("Price ($)")
151. plt.ylabel("Number of Products")
152.
153. plt.grid(axis="y", alpha=0.5)
154.
155. plt.show()
```

Histogram Analysis

Most Products

Price Range:

156. \$1,000 – \$2,500

Small Premium Cluster

Price Range:

157. \$3,500 – \$4,000

Business Insight

SwiftCart mainly sells mid-range products.

However:

High-end products generate strong revenue per unit.

Recommendation:

Launch another premium Home product to capture the luxury market.

ULTIMATE LEVEL HIGHER!

FEATURE ENGINEERING FOR AUTOMATED BUSINESS DECISIONS

Create a recommendation engine.

Business Action Logic

```
158. def recommend_action(row):
159.
160.     if (
161.         row["Total_Sales"] > 10000
162.         and
163.         row["Customer_Rating"] > 4
164.     ):
165.         return "PROMOTE HEAVILY"
166.
167.     elif (
168.         row["Total_Sales"] < 5000
169.         and
170.         row["Customer_Rating"] < 3
171.     ):
172.         return "CONSIDER DISCONTINUE"
173.
174.     else:
175.         return "KEEP AS IS"
```

Apply the Function

```
176. df["Business_Action"] = (
177.     df.apply(
```

```
178.     recommend_action,  
179.     axis=1  
180. )  
181. )  
182.  
183. df[  
184.     [  
185.         "Order_ID",  
186.         "Product_Category",  
187.         "Total_Sales",  
188.         "Customer_Rating",  
189.         "Business_Action"  
190.     ]  
191. ]
```

Business Insight

The dataset now automatically tells management:

- Which products to promote
- Which products to discontinue
- Which products to maintain

This enables data-driven decision-making.

FINAL BUSINESS SUMMARY & RECOMMENDATIONS

1. Inventory Strategy

Increase Home inventory by approximately 40%.

Reason:

- Highest sales
 - Highest ratings
-

2. Marketing Strategy

Feature all Premium Products (Rating > 4) in:

- Email campaigns
 - Website banners
 - Social media advertisements
-

3. Pricing Opportunity

There is a noticeable pricing gap between:

192. \$2,500 – \$3,000

Recommendation:

Launch a new Home product in this range.

4. Books Category Investigation

Books generate the lowest sales.

Potential Causes:

- High pricing
- Low demand
- Weak marketing

Recommendation:

Introduce a promotion:

193. Buy 2 Get 1 Free

to stimulate demand.

TRAINER'S THOUGHT-PROVOKING EXAM QUESTIONS

Question 1

Suppose SwiftCart introduces:

- 10% discount for Clothing
- 20% discount for Electronics

How would you write a `.loc[ ]` condition to update prices?

Question 2

Why is `df.describe()` more useful than simply calculating:

194. `df.mean()`

when making business decisions?

Discuss:

- Minimum values
 - Maximum values
 - Quartiles
 - Distribution insights
-

Question 3

Suppose Home had:

- Highest sales
- Lowest ratings

Would your recommendation remain the same?

Why or why not?

Discuss the trade-off between:

- Revenue
 - Customer satisfaction
 - Long-term sustainability
-

KEY EXAM TAKEAWAY

A Sales Analyst should always follow this workflow:

1. Load Data
2. Explore Data
3. Check Data Quality
4. Calculate Business Metrics
5. Filter Important Segments
6. Sort and Rank Products
7. Create Revenue Metrics
8. Use GroupBy Analysis
9. Visualize Results
10. Make Business Recommendations

Remember:

Data becomes valuable only when it drives business decisions. The goal of Pandas is not just analysis—it is better decision-making.

